COMP 469
Fall 2026

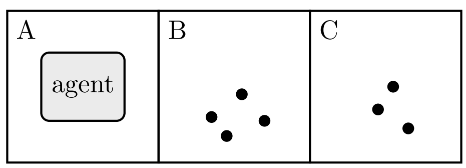
Lab 02

Chapter 2: Intelligent Agents Name(s): ______________________

Covers AIMA 4e §2.1–§2.4.7. **No search algorithm is needed anywhere in this handout.** If you find yourself describing how to *compute* a route, you have wandered into the next chapter — every question here is about what an agent perceives, what counts as success, what kind of environment it faces, and which agent design that environment demands. Work in groups of two or three. Answers are graded on reasoning, not length.

1 The Three-Square Vacuum World

Extend the vacuum world of Figure 2.2 to three squares in a row, A, B, and C. Each square is either Clean or Dirty. The agent perceives its own location and the status of the square it occupies, so a percept looks like $[B, \mathit{Dirty}]$. The available actions are *Left*, *Right*, *Suck*, and *NoOp*. The agent starts in A. Moving into a wall leaves the agent where it is. Sucking cleans the current square, and clean squares stay clean.



Unless a part says otherwise, the performance measure awards one point for each clean square at each time step, over a lifetime of 1000 steps.

(a) How many distinct percepts are there? How many distinct percept sequences of length exactly t? Write a closed-form expression for the number of rows a lookup table needs in order to cover every percept sequence of length 1 through T.

(b) Tabulate the agent function for every percept sequence of length one. Fill in an action that is rational under the performance measure above.

Percept sequence	**Action**
$[A, \mathit{Clean}]$	
$[A, \mathit{Dirty}]$	
$[B, \mathit{Clean}]$	
$[B, \mathit{Dirty}]$	
$[C, \mathit{Clean}]$	
$[C, \mathit{Dirty}]$	

(c) Write a simple reflex agent program, in at most six lines of pseudocode, that produces exactly the table you filled in for part (b). Then evaluate your expression from part (a) at $T = 10$ and compare the two numbers. State in one sentence what the comparison is meant to show.

(d) Now change the performance measure so that each movement costs one point, while *Suck* and *NoOp* are free. Is your agent from part (c) still rational? If not, say what behavior becomes rational once the floor is entirely clean, and name the action that makes it possible.

(e) Your part (c) agent cannot actually tell when the floor is clean, because its action depends only on the current percept. What is the smallest capability you must add, what is the name Chapter 2 gives the resulting design, and which two models does that design maintain?

(f) Suppose the location sensor fails, so the percept is now just *Clean* or *Dirty*. Explain why no simple reflex agent with a deterministic rule set can be rational here. What partial fix does the chapter offer, and why does it call it only a trick rather than a solution?

2 PEAS and Two Task Environments

Consider two systems.

- **Agent I** — a tournament chess program playing under a game clock.
- **Agent II** — an autonomous hospital delivery robot that carries supplies between wards along corridors it shares with people, gurneys, and cleaning carts.

(a) Write a PEAS description for Agent II with at least two concrete items in each category. “Be helpful” and “do a good job” are not performance measures.

Performance	
Environment	
Actuators	
Sensors	

(b) Classify both agents on the seven dimensions of §2.3.2. One or two words per cell.

Dimension	**I: chess with a clock**	**II: hospital robot**
Observable		
Agents		
Determinism		
Episodes		
Time		
State / action		
Knowledge		

(c) Students most often miss the **Time** row. Explain why chess with a clock is semidynamic rather than static, and name a task environment from §2.3.2 that is simply static.

(d) *Known* and *observable* are independent. For the hospital robot, describe (i) a situation in which its environment is known but only partially observable, and (ii) a change that would make some part of it unknown while it remains fully observable.

(e) The chapter separates *nondeterministic* from *stochastic*. Which label applies to the robot's model of a pedestrian stepping into its path, and what would have to change for the other label to apply?

(f) A vendor proposes the performance measure "minimize average delivery time." Give one concrete way a perfectly rational agent could maximize this measure while producing behavior the hospital would reject. Rewrite the measure to close the loophole, and name the problem from Chapter 1 that this illustrates.

3 Agent Designs, Rationality, and Representation

(a) For each system below, select the **least complex** agent design from Chapter 2 that is adequate for the task as described. Choose exactly one bubble per row.

(i) A smoke alarm that sounds whenever particulate density crosses a fixed threshold.

○ simple reflex ○ model-based reflex ○ goal-based ○ utility-based ○ learning

(ii) A floor-cleaning robot whose sensors cannot tell which room it is in, but which must cover every room.

○ simple reflex ○ model-based reflex ○ goal-based ○ utility-based ○ learning

(iii) A delivery drone that must reach a specified address; any route that arrives is equally acceptable.

○ simple reflex ○ model-based reflex ○ goal-based ○ utility-based ○ learning

(iv) A router that must weigh a faster arrival against a higher chance of missing a pickup window.

○ simple reflex ○ model-based reflex ○ goal-based ○ utility-based ○ learning

(v) A mail filter deployed at an organization whose spam patterns were not known at design time.

○ simple reflex ○ model-based reflex ○ goal-based ○ utility-based ○ learning

(b) True or false, with a one-line justification for each.

(i) An agent that chose a rational action and then suffered the worst possible outcome acted irrationally.

(ii) A rational agent never has reason to take an action whose only payoff is information.

(iii) Whether an agent is rational can be determined by inspecting its program.

(iv) An agent that relies entirely on its designer's prior knowledge lacks autonomy.

(c) A table-driven agent stores one entry per percept sequence. Let $|P|$ be the number of distinct percepts and T the agent's lifetime in steps. Write the number of table entries, then give the **three independent** reasons the chapter says this design fails, and name the analogy it offers for what should replace the table.

(d) Match each learning-agent component to its job, then answer the follow-up.

Performance element	
Critic	
Learning element	
Problem generator	

Follow-up: why must the performance standard be fixed and sit conceptually *outside* the agent?

(e) Return to the three-square vacuum of Question 1. Write the state "the agent is in B, A is clean, B is dirty, C is clean" in each of the three representations.

(f) Which of the three representations is required to express "every square to the left of the agent is clean" without enumerating the cases one by one, and what does the chapter say you pay for that expressiveness?

(g) Is a one-hot encoding of the agent's location localist or distributed? Give one advantage of the alternative.

4 Pac-Man as a Chapter 2 Agent

In the two lecture demonstrations, the same simple reflex Pac-Man program was run on an easy corridor level and then on a level with branches and detours. It did well on the first and poorly on the second.

(a) In Chapter 2 vocabulary, what changed between the two runs and what did not? What does this tell you about judging an architecture from one successful run?

(b) A classmate says "it failed because its path-finding routine was too slow." Using the equation *agent = architecture + program*, explain in two or three sentences why this diagnosis is wrong.

(c) Write a performance measure for Pac-Man that rewards survival and nothing else. Describe the degenerate behavior a perfectly rational agent would produce under it, then repair the measure.